Natnael Mola

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Summary

currently a Final year master's student under the EU erasmus mundus joint master's program by autonomous university of madrid, university of bordeaux and pazmany peter catholic university. Have a lot of research experience in machine learning and AI in general, mainly focused in Mechanistic Interpretability, AI safety, Machine unlearning in Vision Models

Education

Erasmus Mundus Joint Master's in Image Processing and Computer Vision with AI (IPCVAI) Sept 2024 – June 2026

Pázmány Péter Catholic University, Autonomous University of Madrid, University of Bordeaux

- **Pázmány Péter Catholic University** (Hungary - 1st semester) GPA: **4.87/5.0**
  - Coursework: Machine Learning Fundamentals, Numerical Analysis, Image Processing, Multi-Sensor Fusion
- **Autonomous University of Madrid** (Spain - 2nd semester) GPA: **4.47/5.0**
  - Coursework: Deep Learning I & II, 3D Vision for Multiple Cameras
- **University of Bordeaux** (France - 3rd semester) GPA: **5.0/5.0**
  - Coursework: Explainable AI, AI and Sustainability

University of electronic science and technology of china Sept 2017 – June 2021

Bachelor's degree in electronic information engineering GPA: **3.72/4.0**

Skills

Proficient in several programming languages

- Python - Advanced
- Matlab - intermediate
- C/C++ - basic

Experiance in deep learning frameworks

- Pytorch - Advanced
- Tensorflow with keras API - intermediate

Experiance with various libraries relevant to machine learning and data science

- Scikit-learn
- Numpy, Pandas
- SAELens, TransformerLens, Prisma

Publications

SPARE: Self-distillation for PARameter-Efficient Removal(arXiv:2602.07058) Submitted to a conference, under review

Natnael Mola, Leonardo S. B. Pereira, Carolina R. Kelsch, Luis H. Arribas, Juan C. S. M. Avedillo

A Deep Learning and Image Enhancement Based Pipeline for Infrared and Visible Image Fusion Neurocomputing, 2024

Jin Qi, Eyob Abera,***Natnael Mola***

Research/Project Experiance

Convex Optimization for distance metric learning in KNN classification

Implemented convex optimization formulation of a distance metric learning problem, and implement solver with projected sub-gradient decent algorithm

Parameter efficient machine unlearning
A research project focusing on finding efficient ways of finetuning state of the art diffusion models to remove unwanted behaviors while preserving general capablities

Analyzing CLIP style text conditioning of Stable Diffusion Models
project focusing on analyzing SD-1.5 which used CLIP for encoding the text prompts. Analyzing which token embeddings from CLIP are the most/least important, analyzing the guidance strength of prompts during the various denoising timesteps, utilizing the CLIP image encoder to do text+image guided generation and other ideas

Current Project

Masters Thesis: Mechanistic Analysis of Knowledge Transfer During Adaptation of Vision Transformers
Using techniques from mechanistic interpretability to understand in depth how models, specifically vision transformers, change during finetuning.

Awards

Betre-Science scholarship - 2017
Scholarship by the Ethiopian government for 100 students from all over the country to study their bachelors abroad with all costs covered

Erasmus Mundus Scholarship in Image processing and computer vision - 2024
EU scholarship that covers all costs for the erasmus mundus joint masters programs in universities across europe